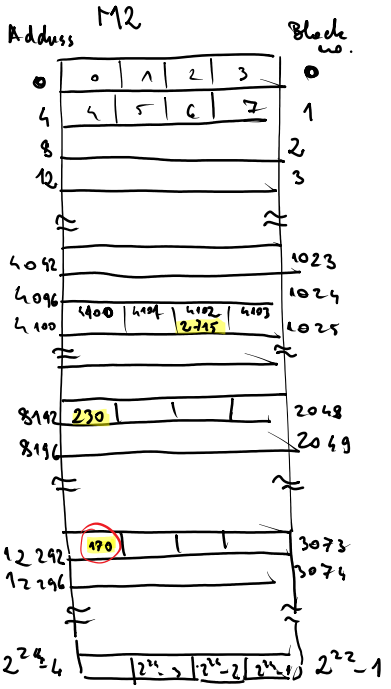
Example 1

- MM – 2^{24} words
- 1 block = 4 words = 2^{2} words
- 1 word = 1B = 2^{0} Bytes
- Cache (data) = 1K blocks = 2^{10} blocks

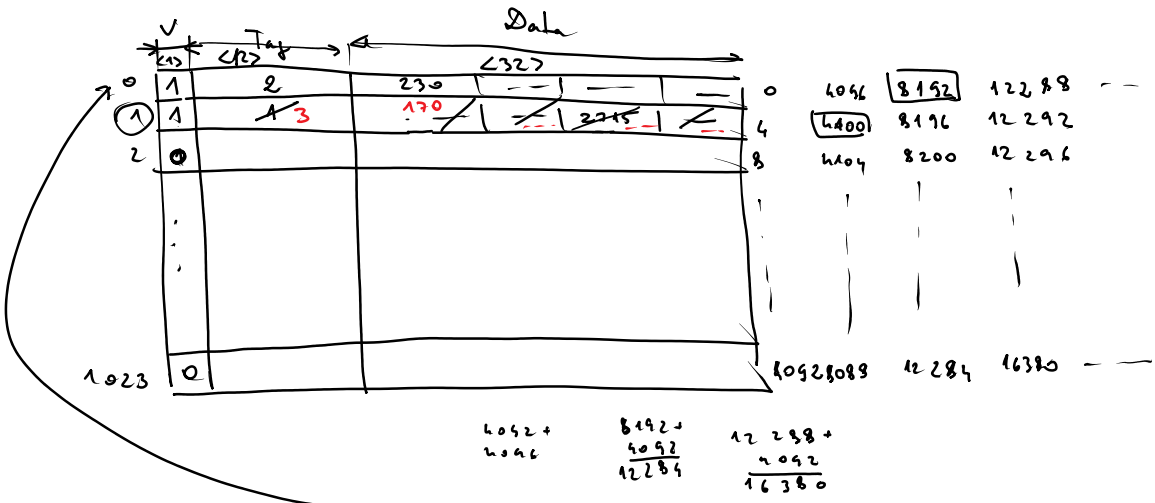
<12>	<10>	<2>
Tag	Index	Block offset

MM Size = 2^{24} words = 2^{24} Bytes = $2^{4} \cdot 2^{20}$ B = 16MiB = 2^{24} words / 2^{2} words/block = 2^{22} blocks/mem

Cache Data Size = 2^{10} blocks = $2^{10} \cdot 2^{2}$ words = 2^{12} words = 2^{12} B = $2^{2} \cdot 2^{10}$ B = 4KiB



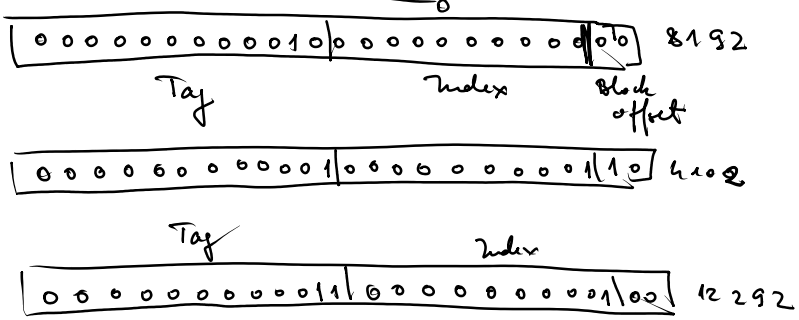
Addr.	Code
8192	230
4102	2715
12292	170



$2^{10} = 1024$
$2^{11} = 2048$
$2^{12} = 4096$
$2^{13} = 8192$

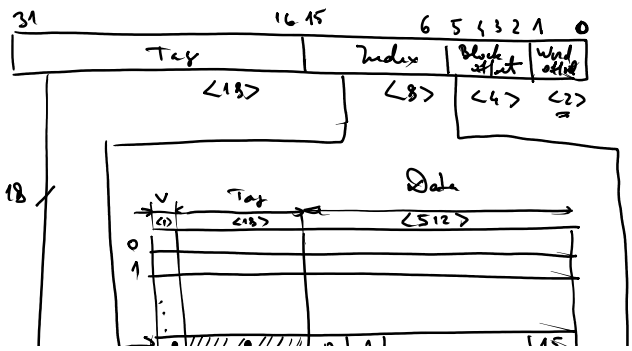
4096 = 2^{12} + 4 = 2^{2}

4096 + 8192 = 12288 + 4 = 12292 $2^{12} + 2^{13} + 2^{2}$

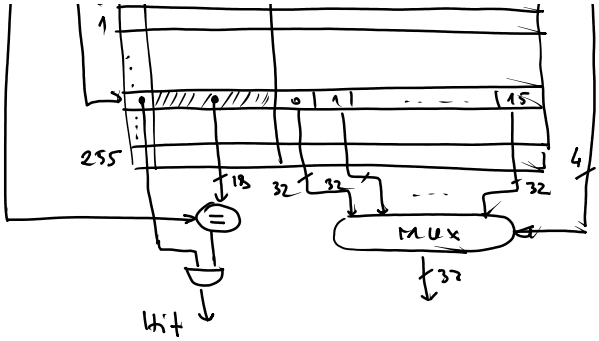


Example 2 Intrinsity FastMATH processor

- embedded processor
- MIPS architecture
- 32-bit address word
- 32-bit data word
- 12-stage pipeline
- separate data & instruction caches
- each cache is 16KiB or 4096 words



- separate data & instruction caches
- each cache is 16 KiB or 4096 words
  - → 16-word blocks ⇒ 2^4 words = 1 block
  - → byte addresses
  - words = 4 Bytes = 2^2 B



$16\,KiB = 2^4 \cdot 2^{10}\,B = 2^{14}\,B$ cache capacity

$1\,\text{block} = 2^4\,\text{words} = 2^4 \cdot 2^2\,B = 2^6\,B$

$$\frac{\text{Cache capacity}}{\text{Block size}} = \frac{2^{14}\,B}{2^6\,B} = 2^8 \text{ blocks / cache}$$

$$MM\ \text{size} = 2^{32}\,B = 2^2 \cdot 2^{30}\,B = 4\,GiB$$

3.2.2 Set-associative mapping

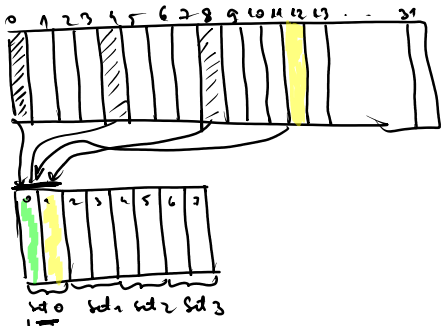
1 cache line = k blocks where $k = 2^s$

$s = 0$ ⇒ Direct Mapping

$s = 1$ ⇒ 2-way SA

$s = 2$ ⇒ 4-way SA

⋮



$M2 \rightarrow 2^m$ blocks

$M1 \rightarrow 2^m$ blocks

$m' = m - s = 2$

$\boxed{m=3,\ s=1}$

$M_1(0), M_1(1), \ldots, M_1(i'), \ldots M_1(2^{m'}-1)$

$m' = m - s$

$i' = j \bmod 2^{m'} = j \bmod 2^{m-s}$

Example 3

- Same as Example 1, only a 2-way SA cache, with the same data size

<13>	<9>	<2>
Tag	Index	Block Offset

Bank 0

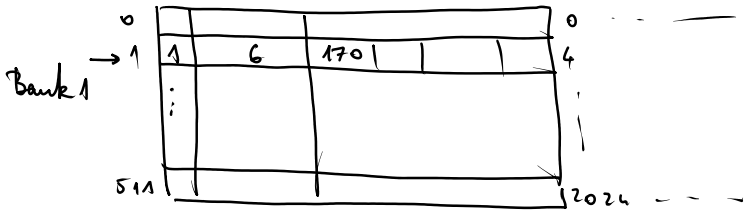
	V <1>	Tag <13>	Data <32>			
0	1	4	230	–	–	–
→ 1	1	2			2715	
511						

0	2048	4096	6144	8192	10240	12288
4	2052	4100	6148	8196	10244	12292
8	2056	4104	6152	8200	10248	12296
2024	4092	6140	8188	10236	12284	

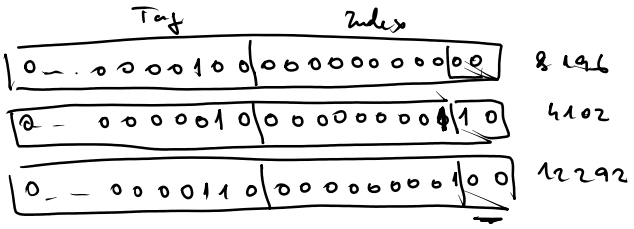
Bank 1

	V <1>	Tag <13>	Data <32>			
0						
→ 1	1	6	170			
4						

Addr.	Code
8192	230



Addr.	Code
8192	230
4102	2715
12292	170



Example 4

VAX 11/780 → 1 MIPS machine

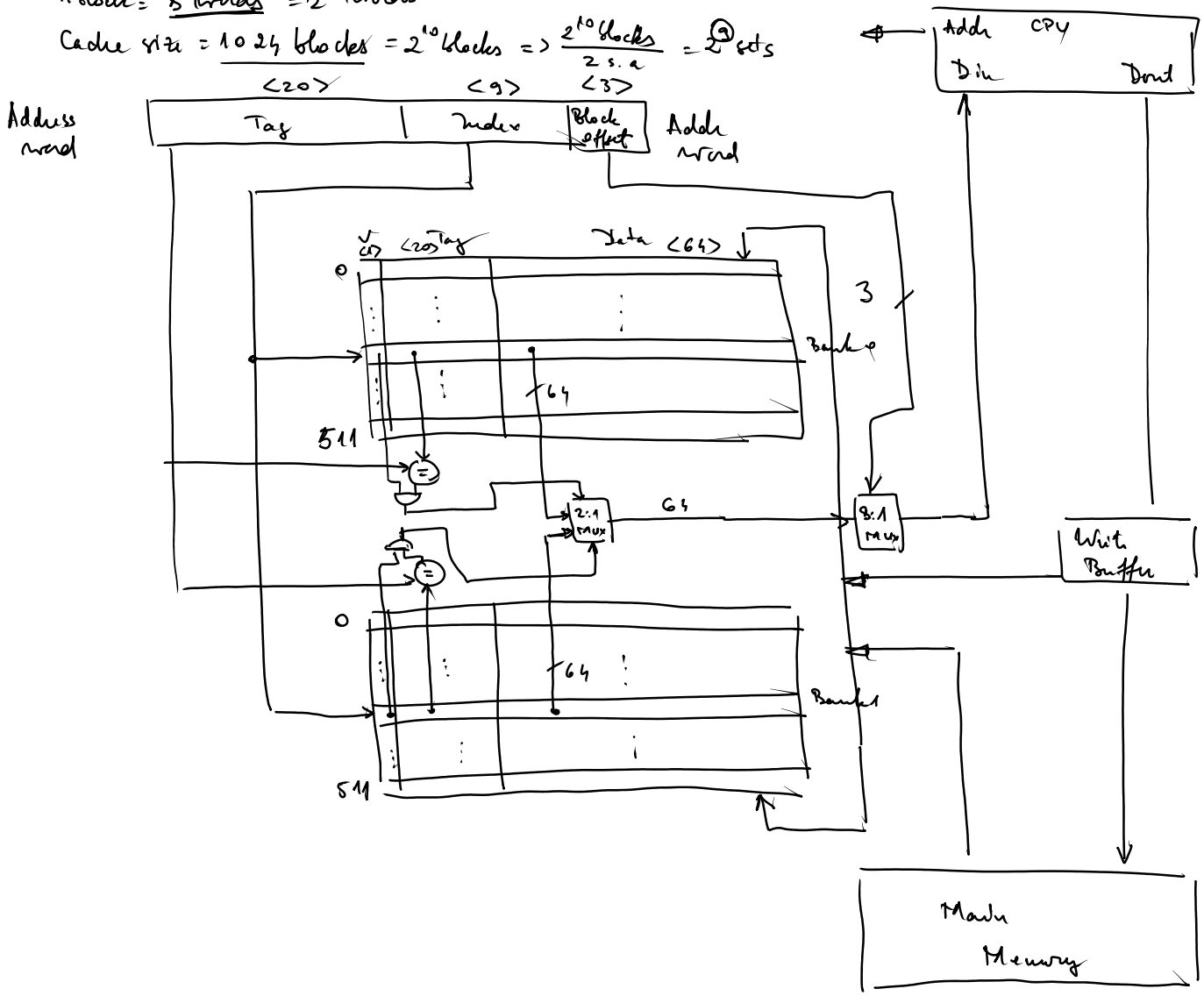
2-way SA

32-bit Address words

1 word = 1 Byte = 2^0 Bytes

1 block = 8 words = 2^3 words

Cache size = 1024 blocks = 2^{10} blocks => $\frac{2^{10}\text{ blocks}}{2\text{ s.a}} = 2^9$ sets



Example 4-way SA cache